MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 1000 VOLTS,

TYPE TSGU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:

Qualification required.

Construction (watertight)

Sizes 3 to 23 inclusive, extruded silicone rubber insulation

First - Copper conductor, uncoated (see Details for size).
Second - Extruded silicone rubber insulation (see Details for diameter).
Third - Glass braid.
Fourth - Braid covering. (See Details for diameter.) Standard Identification Code, applied by method 1.
Fifth - The three conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Either non-fibrous or a combination of non-fibrous and fibrous fillers shall be used to form a firm, well-rounded assembly.
Sixth - Binder or combination binder/barrier.
Seventh - Special thermoplastic jacket (gray)(see Details for thickness). Cable surface marking required.

Sizes 50 to 400, inclusive, silicone rubber-glass tape insulation.

First - Copper conductor, uncoated (see Details for size).
Second - Silicone rubber-glass tape insulation (see Details for diameter). Letter Identification Code, applied by method 5.
Third - The three conductors shall be cabled together with a lay not greater than 24 times the pitch diameter. Either non-fibrous or a combination of non-fibrous and fibrous fillers shall be used to form a firm well-rounded assembly.
Fourth - A binder.
Fifth - Special thermoplastic jacket (gray)(see Details for thickness). Cable surface marking required.

Note 1: When this cable is manufactured with armor, the following additional requirements shall apply:

(B) (a) The type designation shall change to TSGA.
(b) Standard thermoplastic jacket (black or gray) may be used.
(c) Braided metal armor and paint shall be added.
(d) Cable surface marking will not be required.
(B) (e) The maximum overall dimension shall be increased by 0.050-inch.
(f) Group C inspection shall include: Armor - Conformance to material, construction and coverage. When standard thermoplastic jacket is used, the air oven test does not apply.

(B) Denotes changes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Ship Engineering Center, Center Building, SEC 6124, Prince George's Center, Hyattsville, Maryland 20782 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Details

Type and size	Conductor size Navy standard	Diameter over outer insulating tape nominal (nom) (inch)	Diameter over insul. minimum (min) (inch)	Diameter over braid covering maximum (max) (inch)	Jacket thickness nom. (inch)	Overall diameter max (inches)	Conductor resistance per 1000 ft. at 25° Celsius (C) max (ohms)	Insulation resistance per 1000 ft. min (megohms)	Cold bending mandrel diameter max (inches)	Accel-erated service loading (amps)
TSGU-3	3(7)	--	0.096	0.130	0.030	0.411	4.0	500	--	--
TSGU-4	4(7)	--	.112	.143	.030	.449	2.51	500	--	--
TSGU-9	9(7)	--	.154	.187	.040	.575	1.25	500	--	--
TSGU-14	14(7)	--	.230	.262	.040	.720	0.785	500	--	90
TSGU-23	23(7)	--	.284	.316	.050	.870	.496	500	--	115
TSGU-50	50(19)	0.334	--	--	.050	.969	.23	200	12	185
TSGU-75	75(37)	.407	--	--	.050	1.134	.149	175	14	250
TSGU-100	100(61)	.453	--	--	.060	1.266	.114	160	16	295
TSGU-150	150(61)	.557	--	--	.060	1.515	.0718	135	19	400
TSGU-200	200(61)	.634	--	--	.060	1.669	.057	125	21	460
TSGU-300	300(91)	.748	--	--	.075	1.957	.0382	110	24	590
TSGU-400	400(127)	.862	--	--	.075	2.203	.0273	100	27	750

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EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - Ohms/1000 ft., max	(See Details)
Voltage withstand - Volts, root mean square (Vrms), min	
Sizes 3 through 9	3000
Sizes 14 and larger	5000
Insulation resistance - Megohms/1000 ft., min	(See Details)
Group A:	
Visual and dimensional	
Watertightness - See MIL-C-915, table of limits for water leakage	
Abrasion resistance - Sizes 3 through 23, scrapes, min	250
Crack resistance - Sizes 3 through 23	No damage
Group B:	
Gas flame - 1 hour	No failure
Drip - At 95°C ± 1°C	Zero
Cold bending, cable (sizes 50 and larger) (See Details for mandrel diameter)	No damage
Physicals (unaged)	
Insulation (extruded)	
Tensile strength - pound-force per square inch (lbf/in^2), min	750
Elongation - Percent, min	125
Jacket (standard thermoplastic)	
Tensile strength - lbf/in^2, min	1800
Elongation - Percent, min	225
(B) Jacket (special thermoplastic)	
(B) Tensile strength - lbf/in^2, min	2300
(B) Elongation - Percent, min	250
Group C:	
Physicals (aged)	
Jacket (standard thermoplastic)	
Hot oil immersion	
Tensile strength - lbf/in^2, min	1400
Elongation - Percent, min	50
(B) Jacket (special thermoplastic)	
Air oven (168 hours at 136°C)	
(B) Tensile strength - Percent of unaged, min	80
(B) Elongation - Percent of unaged, min	60
(B) Hot oil immersion	
(B) Tensile strength - Percent of unaged, min	75
(B) Elongation - Percent of unaged, min	50
Permanence of printing (conductor) - Sizes 3 thru 23 cycles, min	50
Permanence of printing (jacket) - Cycles, min	250
Cable filler removability	

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical, all Groups A, B and C plus the following:

Cable aging and compatibility (125°C ± 5°C.)	No failure
Cold working (minus 20°C ± 2°C.)	No damage
Gas flame (3 hours)	No failure
Specific gravity of extruded silicone insulation, max	1.55
(B) Accelerated service - (Sizes 14 and larger) (See Details for load current)	No failure

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UNIT ORDERING LENGTHS:

Types and Size	Feet (nom)
TSGU-3	2500
TSGU-4 through -23	2000
TSGU-50	1500
TSGU-75 through -150	1000
TSGU-200 through -400	800

Custodians:
Army - MI
Navy - SH
Air Force - 99
Review activities:
Army - EL, AV
Navy - EC
User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0674-30)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

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